

Homework problems:

Exercise 11.1 (15 points). Solve all the exercises shown on the page Problem H.11.1 - Risk factors for low birth weight until June 29, 09:00.

Remarks:

1. The points for this problem count for the grade bonus. Use the link given on Moodle under the topic "Peer-Review of Homework Problems" to upload your solution.

Your upload consists of the Quarto file containing your solution **and** the HTML file, that you generated by rendering the Quarto file.

Note that you **SHOULDN'T** write your name on your submission. You are identified by signing in on Moodle.

2. You can earn up to **five additional points**. To do this, you need to grade three solutions (your own and two from your peers) that will be assigned to you on Moodle for evaluation. You can gain up to five additional points based on the grading quality. See next week's problem set for more details.

3. Note that there is a template file `starter-low_birth_weight.qmd` on Moodle for this problem (in the folder *Data, Quarto and R files - Sample solutions and previous problem sets*). You can certainly change the name of the file, but don't choose your name as file name.

In case you don't use the template file, make sure to include in the YAML header the option `embed-resources: true`:

```
format:
  html:
    embed-resources: true
```

Solution: Check again the page Problem H.11.1 - Risk factors for low birth weight to find the solution to the problem as well as some **grading instructions**.

Exercise 11.2. The search times $X_1, \dots, X_n$ of n project teams, each tasked with solving the same problem in different companies, can be assumed to be independent and exponentially distributed with parameter θ . Determine the maximum likelihood estimator $\hat{\theta}(X_1, \dots, X_n)$ for θ . Based on the sample `x <- c(x.1,...,x.n)`, calculate the maximum likelihood estimate for θ . For the given sample, the following mean value was obtained.

```
mean(x)
```

```
[1] 98
```

Solution:

Since $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Exp}(\theta)$, the likelihood is given by

$$L(\theta ; \mathbf{x}) = \prod_{i=1}^n f_{\theta}(x_i) = \prod_{i=1}^n \theta e^{-\theta x_i} = \theta^n e^{-\theta \sum_{i=1}^n x_i}.$$

We now obtain the maximum likelihood estimator by maximizing the likelihood or the log-likelihood function

$$\ell(\theta) = \log(L(\theta; \mathbf{x})) .$$

It holds that

$$\ell(\theta) = n \cdot \ln(\theta) - \theta \cdot \sum_{i=1}^n x_i .$$

Differentiating the log-likelihood function and setting it to zero yields

$$\frac{d}{d\theta} \ell(\theta \mid x_1, \dots, x_n) = n \cdot \frac{1}{\theta} - \sum_{i=1}^n x_i \stackrel{!}{=} 0 \quad \Leftrightarrow \quad \hat{\theta} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}_n} .$$

Since

$$\frac{d^2}{d\theta^2} \ell(\theta; \mathbf{x}) = -n \cdot \frac{1}{\theta^2} < 0, \quad \forall \theta \in (0, \infty),$$

then $\hat{\theta}(x_1, \dots, x_n) = \frac{1}{\bar{x}_n}$ is the maximum likelihood estimate and

$$\hat{\theta}(X_1, \dots, X_n) = \frac{1}{\bar{X}_n}$$

is the corresponding maximum likelihood estimator.

For the sample $\mathbf{x}$, this therefore yields an estimate of $\hat{\theta} = \frac{1}{98}$. The expected time to solve the problem is therefore 98 minutes in this fitted model.